Thadomal Shahani Engineering College

Department of Artificial Intelligence and Data Science

Year/Sem: BE/SEM VII **Academic Year: 2026-27**

SUBJECT NAME: BLOCKCHAIN LAB

Lab Outcomes: At the end of the course, the students should be able to

1 Develop and test smart contract on local Blockchain.

2 Develop and test smart contract on Ethereum test networks.

3 Write and deploy smart contract using Remix IDE and Metamask.

4 Design and develop Cryptocurrency.

5 Write and deploy chain code in Hyperledger Fabric.

6 Develop and test a Full-fledged DApp using Ethereum/Hyperledger.

Experiment List

Sr. No.	Experiment Name	LO
1	To study and implement local blockchain using Truffle with integration of Cryptography in Blockchain with Merkle root tree hash	**LO 1**
2	To study and implement Smart contracts and Chain code using Solidity programming language	**LO 1**
3	To study and implement deployment and publishing smart contracts on Ethereum test network	**LO 2**
4	To study and implement Smart contract using Remix IDE and Metamask for development and deployment of Crypto currency	**LO 3** **LO 4**
5	To study and implement the chain code deployment in Hyperledger Fabric	**LO 5**

6	Case Study on Hyperledger Blockchain Platform	**LO 6**
7	Case Study on Bitcoin Blockchain Platform	**LO 4**
8	To study and implement creation and deployment of DAPPs on Ethereum blockchain	**LO 2**
9	Mini-project on Design and Development of a DApps using Ethereum/Hyperledger for commercial applications	**LO 6**

Subject Incharge

[Dr. Bhushan Jadhav]